

DIET INDUCED THERMOGENESIS

Thermogenesis

Thermogenesis is the physiological process of heat production in warm-blooded animals. It is a function that allows organisms to maintain a stable internal body temperature by converting stored chemical energy into heat.

This process can be influenced by several factors, including the type of food consumed, physical activity, and individual metabolic rates.

DIET INDUCED THERMOGENESIS

Diet-induced thermogenesis (DIT) refers to the energy expended by the body to digest, absorb, and metabolize food. This process is a critical part of the thermogenic effect of food, impacting overall energy expenditure significantly. DIT varies according to the macronutrient composition of the diet. For instance:

- **Proteins** typically induce the highest thermogenic response, followed by carbohydrates and then fats.
- The digestion of protein requires more energy due to its complex structure and the processes involved in breaking it down.
- Conversely, fats provide less thermogenic heat since they are simpler to metabolize.

DIET INDUCED THERMOGENESIS

Different foods require different processing energy:

Nutrient

Thermic Effect

Proteins

Highest (~20–30%)

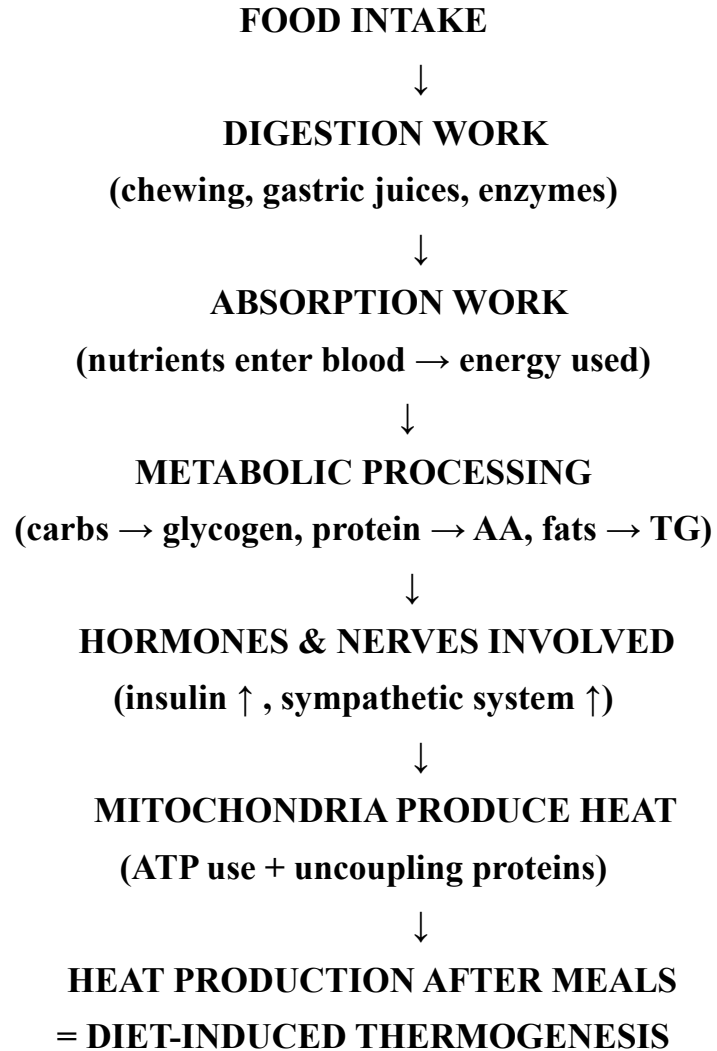
Carbohydrates

Moderate (~5–10%)

Fats

Lowest (~0–3%)

MECHANISM OF DIT



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Diet-induced thermogenesis (DIT) is the increase in energy expenditure that occurs after eating. When food is consumed, the body performs digestion work such as chewing and secretion of digestive juices and enzymes.

After digestion, nutrients are absorbed into the bloodstream, which also requires energy. These absorbed nutrients are further metabolically processed: carbohydrates are stored as glycogen, proteins are converted to amino acids, and fats become triglycerides.

Hormones such as insulin and activation of the sympathetic nervous system enhance this metabolic activity. In the cells, mitochondria use ATP and uncoupling proteins to produce heat. This post-meal heat production is termed diet-induced thermogenesis and contributes to total daily energy expenditure.

SIGNIFICANCE OF DIT

DIT is significant because it:

- ✓ contributes to total energy expenditure
- ✓ influences body weight regulation
- ✓ affects satiety & appetite
- ✓ varies between individuals (metabolic variability)

FACTORS AFFECTING DIT

Factors Affecting DIT

- Meal composition (macro ratio)
- Meal size
- Frequency of eating
- Age
- Gender
- Body composition (muscle $\uparrow$ thermogenesis)
- Hormones (thyroid, catecholamines)
- Physical training status